Hierarchical N-Body Simulation of Galactic Dynamics in WebGPU

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Abstract

GPU-accelerated N-body solvers are well established on native platforms such as CUDA and Metal, but no prior work has implemented, optimised, and systematically benchmarked a hierarchical Barnes–Hut solver in WebGPU, which is the only current GPU API that combines compute shaders with browser deployability. We evaluate WebGPU as a compute platform for scientific simulation by implementing and benchmarking a complete Barnes–Hut gravitational N-body solver. The solver constructs and traverses a Linear Bounding Volume Hierarchy (LBVH) entirely on the GPU each timestep using the parallel method of Karras, combined with a symplectic leapfrog integrator. A single C++/WGSL codebase targets both native desktop execution and browser deployment via WebAssembly. We characterise performance across four WebGPU implementations (wgpu-native, Dawn, Chrome, and Safari) on an Apple M2 and compare against a native Metal Barnes–Hut baseline to isolate the cost of the WebGPU abstraction layer. At $N = 1000$ the abstraction layer imposes a 2.0 $\times$ overhead relative to native Metal; at $N \geq 5000$ the WebGPU solver outperforms the baseline, reaching 2.9$\times$ faster at $N = 100000$. Force evaluation accounts for 94–99% of total step time, with LBVH construction staying below 0.35 ms at all tested particle counts. Browser execution via Emscripten introduces a scaling overhead that narrows from 2.0$\times$ to 1.4$\times$ as GPU compute dominates, preserving the qualitative scaling behaviour seen in native execution. The 32-bit floating-point precision of current WebGPU implementations, together with the tree force approximation, constrains numerical fidelity. The results show that WebGPU's compute shader model can support a hierarchical N-body solver with parallel tree construction, and that the abstraction overhead is low enough for interactive galactic dynamics simulation without specialised hardware or vendor-locked APIs. These findings establish a performance baseline for GPU-accelerated scientific computing in WebGPU and demonstrate that browser-based deployment of computationally demanding simulations is now practical.

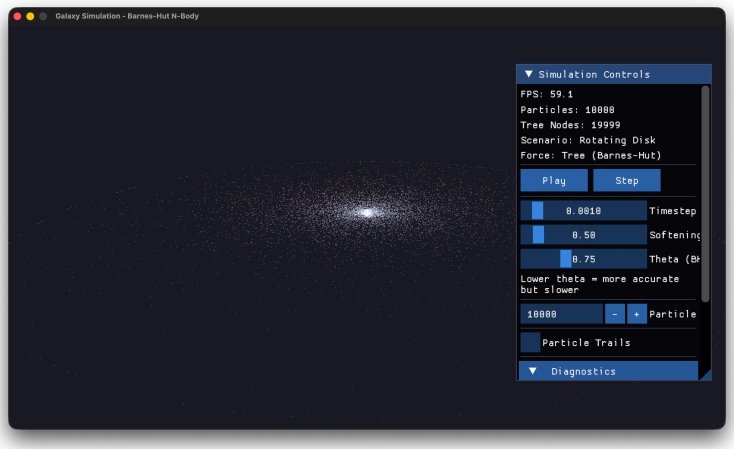

Figure 1: The interactive simulator running a rotating exponential disk.

How It Works

Each timestep, particles are advanced by the leapfrog integrator, then the entire tree is rebuilt on the GPU: positions are hashed into Morton codes (a space-filling Z-curve), radix-sorted so nearby particles sit adjacent in memory, and assembled into a binary bounding volume hierarchy using Karras's parallel algorithm. Every particle then walks the tree top-down — accepting distant nodes as single-mass approximations and expanding nearby ones — reducing the cost from $O(N^2)$ to roughly $O(N \log N)$. All six construction passes and the force traversal run on the GPU with no CPU involvement per step.

Key Results

Almost all of the GPU's time goes into traversing the tree to compute forces — **94–99% of every timestep**, depending on particle count. Building the tree itself is cheap: the six-pass LBVH pipeline finishes in under a third of a millisecond even at 100K particles. If there is performance left on the table, it is in the traversal shader.

WebGPU vs Native Metal

The most surprising result is that **our WebGPU solver beats a native Metal implementation** once the workload is large enough. We tested against UniSim, an open-source Metal Barnes–Hut code running on the same Apple M2 with the same GPU driver. At small particle counts WebGPU pays a dispatch-overhead tax, but by $N = 5000$ the fully GPU-resident tree pipeline pulls ahead — and the gap keeps widening.

N	Metal (ms)	WebGPU (ms)	Ratio
1,000	2.94	5.86	2.0× slower
5,000	10.09	7.35	**1.4× faster**
10,000	21.35	11.00	**1.9× faster**
50,000	121.77	65.46	**1.9× faster**
100,000	516.78	180.11	**2.9× faster**

Table 1: WebGPU (wgpu-native) vs native Metal (UniSim) on Apple M2.

Implementation choice matters

Not all WebGPU backends are equal. We ran the same solver on four implementations — all using the same Metal driver on the same machine — and saw **large performance differences**. Which backend is fastest depends on the workload size: Dawn wins at small dispatches, wgpu-native wins at scale. In the browser, Chrome is actually faster than native wgpu at low particle counts but scales worse, levelling off at roughly **1.4× overhead** for large simulations.

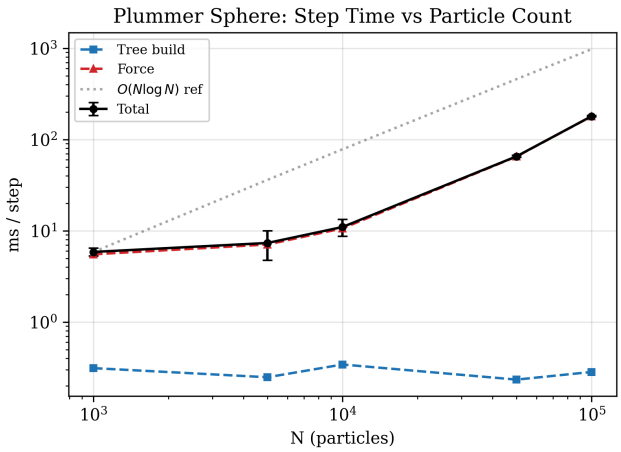

Figure 2: Runtime vs particle count (log-log).

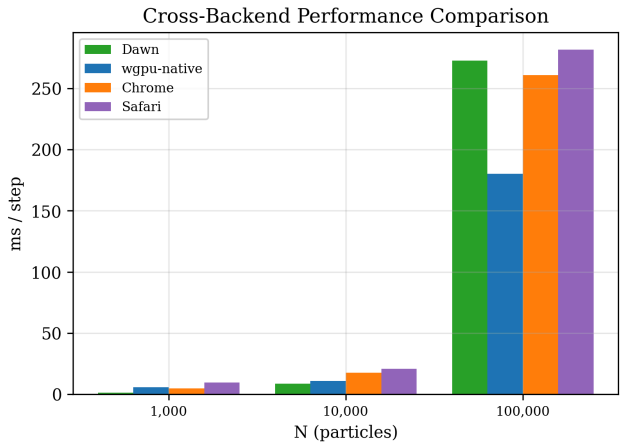

Figure 3: Four WebGPU implementations compared.